

Problem Statement and Goals

Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
September 24	Robert	Write problem statement and inputs/outputs
Date2	Name(s)	Description of changes
...	...	...

1 Problem Statement

[You should check your problem statement with the [problem statement checklist](#) —SS].

[You can change the section headings, as long as you include the required information. —SS]

1.1 Problem

Current AI inference markets are increasingly concentrated around a small number of large organizations [?]. As the artificial intelligence (AI) industry continues to grow, we become increasingly reliant on centralized providers for AI models, data, hardware, and compute. This centralization increases the risk of biased outputs because of the lack of diversity in training algorithms and data used by these companies [?]. Furthermore, the oligopolized nature of the AI market creates a risk of censorship and manipulation of AI outputs. The lack of transparency from these companies and the black-box nature of many AI models makes it difficult for consumers to verify the quality and reliability of their inferences. Lastly, the centralized nature of the compute infrastructure

used by these companies creates potential single points of failure. According to the 2026 Stanford AI Index, NVIDIA accounts for over 60% of total global AI compute capacity [?]. This concentration poses a significant risk of disruption to AI markets in the event of a single failure.

[What problem are you trying to solve? Why is it important? Do not talk about how you will solve the problem, only what the problem is. AI is rarely part of the problem. The problem is “what” you want to do, not “how” you want to do it. The problem statement should be written in a way that is understandable to a non-technical audience. —SS]

1.2 Inputs and Outputs

The inputs to the system consists of task definitions, inference requests, and network participation. Task definitions are specification of a problem to be solved via machine learning, including the expected input and output types. Inference requests ask the network to perform inference for a given task and include input data or some specification of where to find the input data. Network participation includes machine learning predictions for individual inference requests that are supplied to the network by an individual.

The system outputs aggregated inference results and maintains information about the network. The aggregated inference results include the final prediction for a given inference request, as well as information about the quality of the prediction.

[Characterize the problem in terms of “high level” inputs and outputs. Use abstraction so that you can avoid details. —SS]

1.3 Stakeholders

- Task requesters
- Inference consumers
- Inference providers
 - Model owners
developers
 - Model runners
- Data providers
- TEST

1.4 Environment

[Hardware and Software Environment for the users. The developer environment is summarized as part of the developer plan. —SS]

2 Goals

This project has a total of five primary goals.

- 2.1 **Automated Decentralized Execution of Transactions and Agreements:** Develop a system that enables the automated and decentralized execution of marketplace transactions and agreements between users and model providers.
- 2.2 **Global AI Model Marketplace:** Develop a global marketplace that connects users with AI models, allowing models to be discovered and ranked based on their reputation.
- 2.3 **Linear Regression Inference Support:** Support linear regression inference tasks, allowing users to submit inference requests to available models through the decentralized marketplace.
- 2.4 **Consumer-Side User Interface:** Develop an intuitive consumer-side interface that allows users to interact with the marketplace and submit inference requests without requiring knowledge of the underlying decentralized infrastructure.
- 2.5 **Decentralized Backend Communication for Model Providers:** Enable model providers to communicate with and interact with the decentralized backend in order to register their models, receive inference requests, and provide inference results.

3 Stretch Goals

This project has a total of 2 stretch goals.

- 3.1 **Support for More Complex Inference Tasks:** Extend the system beyond linear regression to support more complex inference tasks and AI models.
- 3.2 **Private Inference Using Trusted Execution Environments:** Investigate and implement Trusted Execution Environments (TEEs) to enable inference tasks to be executed on completely private data while protecting the data from unauthorized access.

[Goals you hope to get to, but not necessary. They are also helpful if the scope of the original project needs to be expanded. —SS]

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?